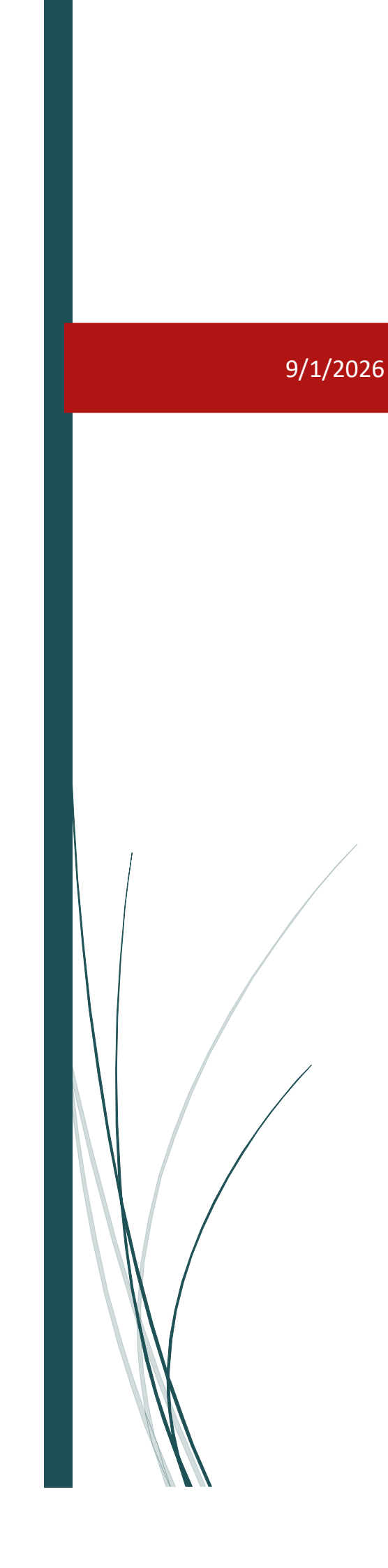


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9/1/2026

N + - Network Fundamentals & Building Networks.

Introduction to Network Devices.

TASK 1.

❖ Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

➤ Network Devices and Typical Ports:

Network Device	Main Function	Typical Ports Used
Switch	Connects multiple wired devices within a local network (LAN).	Ethernet (RJ-45)
Router	Connects the home network to the Internet and routes data between networks.	WAN and LAN Ethernet (RJ-45)
Hub	Connects multiple wired devices and broadcasts data to all connected devices.	Ethernet (RJ-45)
Firewall	Monitors and filters network traffic to block unauthorized access.	Ethernet (RJ-45), WAN/LAN
Access Point	Provides Wi-Fi connectivity to wireless devices such as phones and laptops.	Ethernet (RJ-45) + Wi-Fi

TASK 2.

❖ Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

➤ Network Devices and Their Ports:

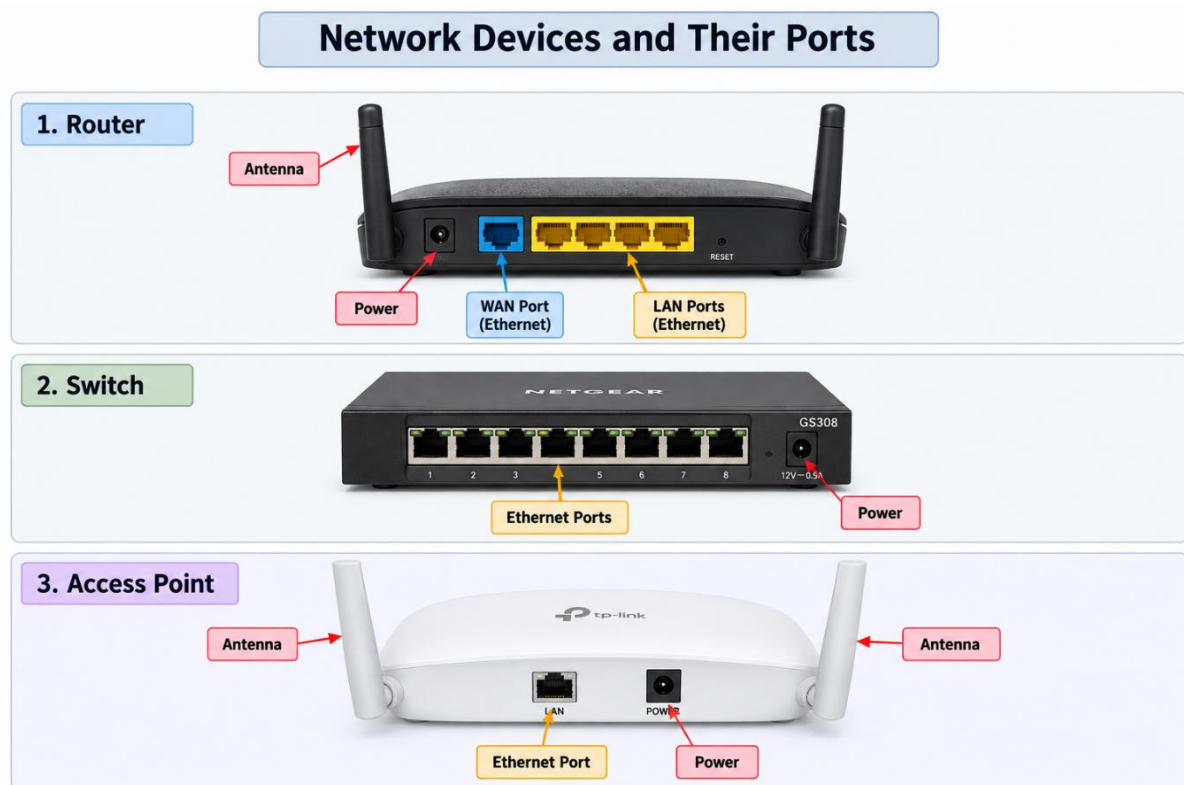


Figure : Network Devices and Their Ports.

The figure shows three network devices: **Router, Switch, and Access Point**, with their important ports labelled.

- ⌘ **Router:** WAN/LAN Ethernet, Power, Antenna
- ⌘ **Switch:** Ethernet Ports, Power
- ⌘ **Access Point:** Ethernet, Power, Antenna
- ⌘ **Console Port:** Not shown in the given diagrams.

TASK 3.

- ❖ Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

- **Role of Network Devices in Zomato:**

I use **Zomato** to browse restaurants and order food online.

The **Access Point** connects my phone to the Wi-Fi network and sends my request to the local network.

The **Router** forwards the request from my home network to Zomato's servers over the Internet.

A **Firewall** helps protect the connection by filtering unauthorized or suspicious network traffic.

TASK 4.

- ❖ Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help, Think about device capacity, coverage, and traffic handling.

- **College Wi-Fi Performance Issue:**

One possible reason for the slow and unstable college Wi-Fi is that the **Access Point is handling too many devices at the same time**, which can exceed its capacity and reduce performance.

Upgrading to a **higher-capacity Access Point** can support more users and handle network traffic more efficiently.

If the problem is due to weak coverage, additional Access Points can also improve the Wi-Fi signal in poorly covered areas.